

Module 4: Apriori & FP-Growth Pattern Mining

Support/Confidence, Apriori Join & Prune, FP-Tree Mining & Lift

Module IV: Frequent Pattern & Association Rule Mining — Complete 7-Topic Tracker

Topic 30: Basic Concepts of Frequent Pattern Mining

Topic 32: Correlation Measures (Lift & Chi-Square Analysis) [UPLOADED PYQ]

Topic 34: The Apriori Algorithm (Join, Prune & Improvements) [UPLOADED PYQ]

Topic 36: Interesting Pattern Evaluation Measures (S-C-L Triad)

Topic 31: Association Rules (Support & Confidence Metrics)

Topic 33: Frequent Itemset Mining Methods Overview

Topic 35: The FP-Growth Algorithm (FP-Tree Construction & Mining) [UPLOADED PYQ]

Topic 30 & 31: Association Rule Mining Formulations

1. Support and Confidence Formulations for Rule $A \Rightarrow B$:

$$\text{Support}(A \Rightarrow B) = P(A \cup B) = \frac{\text{Count}(A \cup B)}{|D|}$$

$$\text{Confidence}(A \Rightarrow B) = P(B | A) = \frac{\text{Support}(A \cup B)}{\text{Support}(A)} = \frac{\text{Count}(A \cup B)}{\text{Count}(A)}$$

A rule is **Strong** if $\text{Support} \geq \text{min_sup}$ and $\text{Confidence} \geq \text{min_conf}$.

Topic 32: Correlation Measures & The Lift Metric [UPLOADED PYQ]

Lift Correlation Formula [UPLOADED PYQ — IT426 Mid]:

$$\text{Lift}(A \Rightarrow B) = \frac{P(A \cup B)}{P(A) \cdot P(B)} = \frac{\text{Confidence}(A \Rightarrow B)}{\text{Support}(B)}$$

- **Lift > 1:** Positive correlation (A and B occur together more frequently than expected by chance).
- **Lift = 1:** Statistical independence (no correlation between A and B).
- **Lift < 1:** Negative correlation (the presence of A discourages the occurrence of B).

Topic 34: The Apriori Algorithm (Join & Prune Steps) [UPLOADED PYQ]

The Apriori Property & Pruning Theorem (Agrawal & Srikant, 1994)

Apriori Invariant: All non-empty subsets of a frequent itemset must also be frequent.

Anti-Monotonicity Pruning Rule: If an itemset is infrequent, all of its supersets are guaranteed to be infrequent and can be immediately pruned without scanning the database!

1. **Join Step:** C_k candidate itemsets are generated by joining L_{k-1} with itself ($L_{k-1} \bowtie L_{k-1}$).
2. **Prune Step:** Any candidate $c \in C_k$ is deleted if any of its $(k-1)$ -subsets is not present in L_{k-1} .

Topic 35: The FP-Growth Algorithm [UPLOADED PYQ]

FP-Growth (Frequent Pattern Growth) overcomes the major computational bottlenecks of Apriori (repeated database scans and massive candidate generation) by constructing a compact prefix-tree structure called the **FP-Tree**.

 **[UPLOADED PYQ — IT335 End] Full Solved Numerical: FP-Growth on Transaction Database ($\text{min_sup} = 60\% = 3$ out of 5)**

Given 5 Transactions:

TID	Items Purchased	Ordered Frequent Items (Support ≥ 3)
10	Beer, Nuts, Diapers	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4 \rangle$
20	Beer, Coffee, Diapers, Nuts	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4 \rangle$ (Coffee dropped)
30	Beer, Diapers, Eggs	$\langle \text{Beer} : 4, \text{Diapers} : 4, \text{Eggs} : 3 \rangle$
40	Beer, Nuts, Eggs, Milk	$\langle \text{Nuts} : 4, \text{Beer} : 4, \text{Eggs} : 3 \rangle$ (Milk dropped)
50	Nuts, Coffee, Diapers, Eggs, Milk	$\langle \text{Nuts} : 4, \text{Diapers} : 4, \text{Eggs} : 3 \rangle$

Step 1: Frequent 1-Itemsets: $\{\text{Nuts} : 4, \text{Beer} : 4, \text{Diapers} : 4, \text{Eggs} : 3\}$. Frequency Order: $\text{Nuts} \rightarrow \text{Beer} \rightarrow \text{Diapers} \rightarrow \text{Eggs}$.

Step 2: Mining Conditional Pattern Bases from Bottom of Header Table:

- **Item `Eggs`:** Conditional Pattern Base: $\{\langle \text{Beer}, \text{Diapers} : 1 \rangle, \langle \text{Nuts}, \text{Beer} : 1 \rangle, \langle \text{Nuts}, \text{Diapers} : 1 \rangle\}$. Conditional FP-Tree: $\emptyset$ (No item has support ≥ 3). Frequent Itemset: $\{\text{Eggs} : 3\}$.
- **Item `Diapers`:** Conditional Pattern Base: $\{\langle \text{Nuts}, \text{Beer} : 2 \rangle, \langle \text{Beer} : 1 \rangle, \langle \text{Nuts} : 1 \rangle\}$. Conditional FP-Tree: $\langle \text{Nuts} : 3, \text{Beer} : 3 \rangle$. Frequent Itemsets: $\{\text{Diapers}, \text{Nuts} : 3, \text{Diapers}, \text{Beer} : 3\}$.
- **Item `Beer`:** Conditional Pattern Base: $\{\langle \text{Nuts} : 3 \rangle\}$. Conditional FP-Tree: $\langle \text{Nuts} : 3 \rangle$. Frequent Itemset: $\{\text{Beer}, \text{Nuts} : 3\}$.

M4 Active Recall & Exam Questions [UPLOADED PYQ]

Q1. [UPLOADED PYQ — IT426 Mid] Compare the significance of Support and Confidence in association analysis.

Why is Lift necessary? (8 Marks)

- **Support:** Measures how frequently an itemset appears across the entire dataset ($P(A \cup B)$). It filters out statistically insignificant random noise.
- **Confidence:** Measures the conditional reliability of the inference rule ($P(B | A)$). High confidence means B is likely when A occurs.
- **Why Lift is Necessary:** Support and Confidence alone can be deceptive. For example, if 90% of all customers buy Tea ($P(\text{Tea}) = 0.9$) and rule `Coffee => Tea` has 80% confidence, Coffee actually *decreases* the likelihood of buying Tea! Lift computes $\frac{0.80}{0.90} = 0.89 < 1$, exposing the negative correlation that Confidence hid.